

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

Mid-Semester Examination, Spring 2025

B. Tech. 6th Semester

Subject Name: Computer Networks

Subject Code: CS3002

Number of pages: 2

Maximum Marks: 30

Exam Duration: 2 Hours

Note: There are **Five** questions. Attempt **All** the questions. All parts of the same question must be attempted in a sequence.

Q.1 (a) List three key functions of *Network Layer* and *Data Link Layer* each.

- (b) What layer in the IP stack best corresponds to the phrase: 'moves datagrams from the source host to the destination host'. Justify your answer.
- (c) Machine X and Y are each connected to a switch S via 100-Kbps link. The propagation delay on each link is 100 ms. S is a store-and-forward device; it begins retransmitting a received packet 20 ms after it has finished receiving it. Calculate the total time required to transmit 20,000 bits from X to Y.

[2+2+2=6 Marks, CO1 (L2)]

Q.2 Consider the two scenarios below:

Scenario 1: A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 150 Mbps.

Scenario 2: A packet-switching scenario with N_{ps} users sharing a 150 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 10 percent of the time.

Answer the following based on the Scenarios mentioned above:

- (a) When circuit switching is used, what is the maximum number of users that can be supported?
- (b) Suppose packet switching is used. If there are 11 packet-switching users, can this many users be supported under circuit-switching? Yes or No.
- (c) Suppose packet switching is used. If there are 20 packet-switching users. What is the probability that at least 2 users are transmitting simultaneously?

[2+2+2=6 Marks, CO1 (L3)]

Q.3 Suppose you open a startup company “Qmap” and want to set up your company network. Your network has the following servers:

- DNS server: “dns1.qmap.com” with IP as “168.119.17.8”
- Web server: “qmap.com” with two IP as “168.119.17.33” and “168.119.17.34”. The web server also has a name as “www.qmap.com”.
- Email server: “mail.qmap.com” with IP as “168.119.17.16” Your Company’s email address is “xyz@qmap.com”.

- (a) What resource records (RRs) do you need to provide to the upper-level “.com” Registrar?
- (b) What RRs do you need to put in your company’s DNS server?

[3+3=6 Marks, CO2 (L3)]

Q.4 (a) Why web-caching is needed? List out the alternative of web-caching.

(b) Explain the working of SMTP Protocol with the help of a Diagram. How's it different from HTTP?

(c) Answer and Justify: Whether HTTP and FTP both are in band or not?

[2+2+2=6 Marks, CO2 (L2)]

Q.5 Suppose that TCP's current estimated values for the round trip time (*estimatedRTT*) and deviation in the RTT (*DevRTT*) are 250 msec and 30 msec, respectively. Suppose that the next three measured values of the RTT are 290 msec, 260 msec, and 310 msec respectively. Compute TCP's new value of *DevRTT*, *estimatedRTT*, and the *TCP timeout* value after each of these three measured RTT values is obtained. Use the values of $\alpha = 0.125$, and $\beta = 0.25$.

[2+2+2=6 Marks, CO3 (L4)]